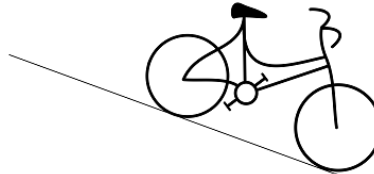


Lecture 10 – Dynamics of rolling motion

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In this lecture we consider how the **rolling motion** of a **rigid body** changes in the course of time.

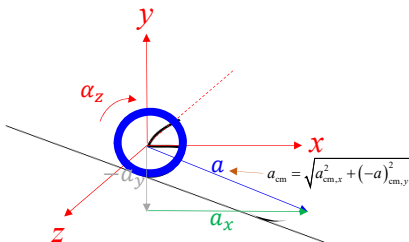


A few slides in this chapter come from Chapter-10 of Young's textbook.

This PPT file will well define the scope of your tests.

1

Rigid-body rotation about a moving axis



A **rolling rigid body** can be considered as a combination of **translational motion of the centre of mass** and **rotation about an axis through the centre of mass**:

$$F_x = Ma_{cm,x}, \quad F_y = Ma_{cm,y},$$

$$\tau_z = I_{cm}\alpha_z.$$

Here, (F_x, F_y) is the net (external) force acting on the rigid body. M is the mass of the rigid body. $(a_{cm,x}, a_{cm,y})$ is its centre-of-mass acceleration, where $a_{cm}^2 = a_{cm,x}^2 + a_{cm,y}^2$. τ_z is the net (external) torque, I_{cm} is the moment of inertia and α_z is the angular acceleration about the axis through the centre of mass of the rigid body.

In particular, the total kinetic energy of a **rolling rigid body** is

$$KE_{\text{total}} = \frac{1}{2}Mv_{cm}^2 + \frac{1}{2}I_{cm}\omega^2.$$

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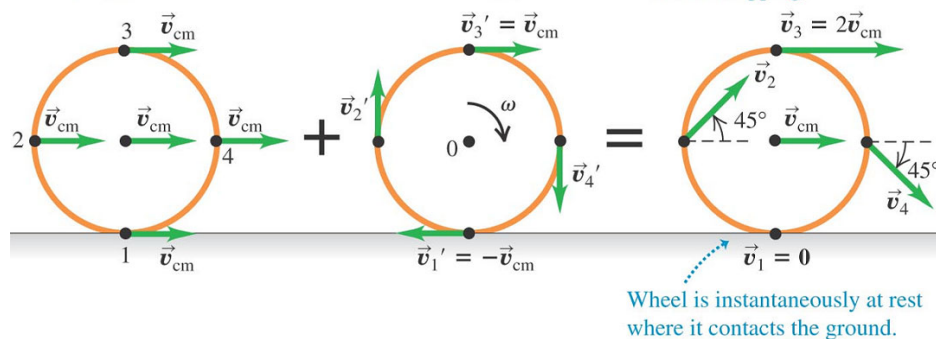
Rigid-body rotation about a moving axis (cont.)

Remark – pure rolling or rolling without slipping.

Translation of the center of mass of the wheel: velocity $\vec{v}_{cm}$

Rotation of the wheel around the center of mass: for rolling without slipping, the speed at the rim must be v_{cm} .

Combination of translation and rotation: rolling without slipping



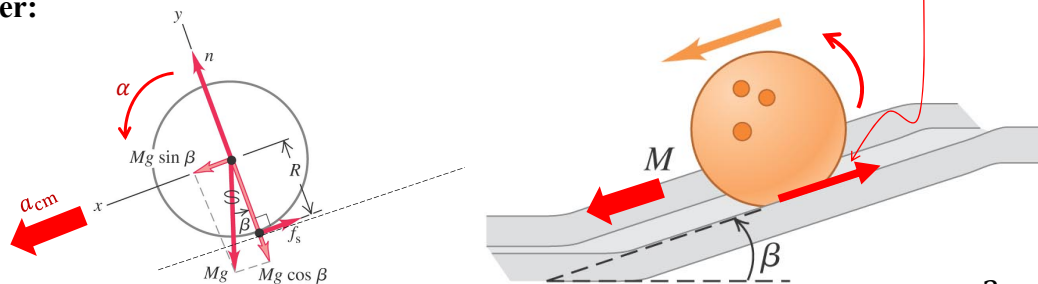
$$v_{cm} = R\omega, \quad a_{cm} = R\alpha.$$

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Example 1.

A bowling ball of radius R rolls without slipping down a ramp, which is inclined at an angle β to the horizontal. What are the ball's acceleration and the magnitude of the friction force on the ball?

Answer:



Treating the ball as a uniform solid sphere, i.e., ignoring the finger holes, $I_{cm} = \frac{2}{5}MR^2$.

By Newton's equations of motion,

$$Mg \sin \beta - f_s = Ma_{cm}, \quad Mg \cos \beta - n = 0, \quad f_s \times R \times \sin 90^\circ = \left(\frac{2}{5}MR^2\right)\alpha.$$

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Example 1. (cont.)

Analysis by Newton's 2nd Law:

By Newton's equations of motion,

$$Mg \sin \beta - f_s = Ma_{\text{cm}} \dots (1) \quad Mg \cos \beta - n = 0 \dots (2) \quad f_s R = \left(\frac{2}{5} MR^2 \right) \alpha \dots (3)$$

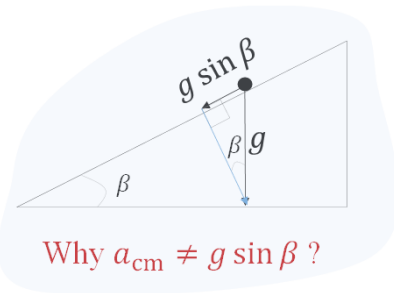
Since the ball **rolls without slipping**, $a_{\text{cm}} = R\alpha$.

$$\text{From (3), } f_s = \frac{2}{5} M(R\alpha) = \frac{2}{5} Ma_{\text{cm}}.$$

$$(1) \rightarrow Mg \sin \beta - \frac{2}{5} Ma_{\text{cm}} = Ma_{\text{cm}} \rightarrow Mg \sin \beta = \frac{7}{5} Ma_{\text{cm}}$$

$$\rightarrow a_{\text{cm}} = \frac{5}{7} g \sin \beta$$

$$\text{So, } f_s = \frac{2}{5} Ma_{\text{cm}} = \frac{2}{5} M \left(\frac{5}{7} g \sin \beta \right) = \frac{2}{7} Mg \sin \beta.$$



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Example 1. (cont.)

Analysis by Energy:

By conservation of total mechanical energy,

$$Mgh = \frac{1}{2} Mv_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$$

Since the ball **rolls without slipping**,

$$v_{\text{cm}} = R\omega \rightarrow \omega = \frac{v_{\text{cm}}}{R}$$

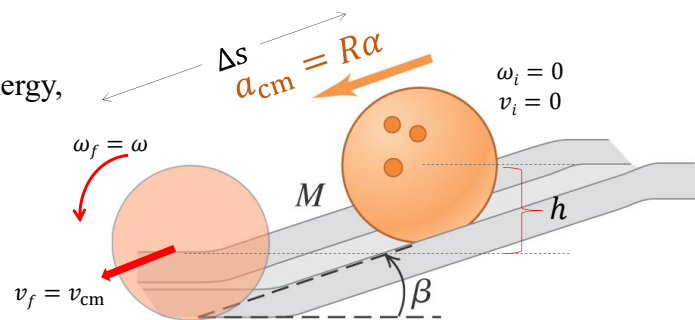
Therefore,

$$Mgh = \frac{1}{2} Mv_{\text{cm}}^2 + \frac{1}{2} \left(\frac{2MR^2}{5} \right) \left(\frac{v_{\text{cm}}}{R} \right)^2 \rightarrow gh = \frac{7}{10} v_{\text{cm}}^2 \rightarrow v_{\text{cm}}^2 = \frac{10}{7} gh.$$

By kinematic equation $v_f^2 = v_i^2 + 2a\Delta s$ where $\sin \beta = \frac{h}{\Delta s}$

$$\frac{10}{7} gh = 2a \left(\frac{h}{\sin \beta} \right) \rightarrow a_{\text{cm}} = \frac{5}{7} g \sin \beta.$$

Same answer.



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Example 1. (cont.)

To find the magnitude of the static frictional force on the ball, we have to invoke Newton's equation of motion

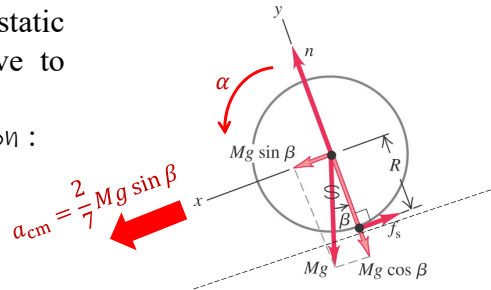
Net force = mass $\times$ acceleration :

$$Mg \sin \beta - f_s = Ma_{\text{cm}}$$

$$\rightarrow f_s = Mg \sin \beta - Ma_{\text{cm}}$$

$$= Mg \sin \beta - M \left(\frac{5}{7} g \sin \beta \right)$$

$$= \frac{2}{7} Mg \sin \beta \quad \text{Same answer.}$$



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Problem. A vertical stick of 1.00 m long and with a mass of 15.0 kg can rotate freely about a hinge at ceiling. A bullet with a mass of 10.0 g and a speed of 400 m/s strikes the centre of the stick, in a direction perpendicular to the stick, and embeds itself there. Find the vertical displacement for the stick (with bullet).

Answer:

Initial angular momentum in counter-clockwise direction:

$$I\omega_i = (m\ell^2) \left(\frac{v_{\text{bullet}}}{\ell} \right) = m\ell v_{\text{bullet}}$$

By conservation of angular momentum, $I\omega_i = I_{\text{total}}\omega_f$.

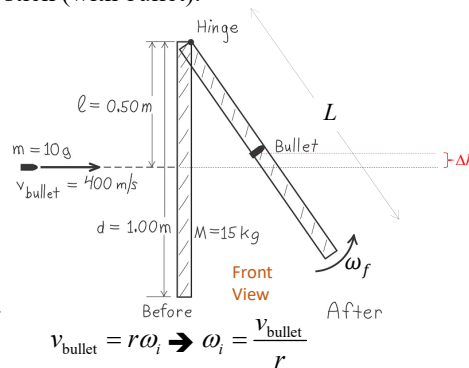
$$\rightarrow m\ell v_{\text{bullet}} = \left(m\ell^2 + \frac{1}{3}ML^2 \right) \omega_f = \left(m\ell^2 + \frac{1}{3}M(2\ell)^2 \right) \omega_f$$

$$\rightarrow 0.01 \times 0.5 \times 400 = \left(0.01 \times 0.5^2 + \frac{1}{3} \times 15 \times (2 \times 0.5)^2 \right) \omega_f$$

$$\rightarrow \omega_f = 0.400 \text{ rad/s}$$

$$\text{Final rotational KE} = \frac{1}{2} I_{\text{total}} \omega_f^2 = \frac{1}{2} \left(m\ell^2 + \frac{1}{3}ML^2 \right) \omega_f^2 = \frac{1}{2} \left(0.01 \times 0.5^2 + \frac{1}{3} \times 15 \times 1^2 \right) 0.4^2 = 0.400 \text{ J}$$

$$(m_1 + m_2)g\Delta h = 0.400 \text{ J} \rightarrow \Delta h = \frac{0.4 \text{ J}}{(15 \text{ kg} + 0.01 \text{ kg})(9.8 \text{ m/s}^2)} = 2.72 \times 10^{-3} \text{ m} = 2.72 \text{ mm}$$



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Example 2.

In a Physics demonstration, your professor “races” various bodies that roll without slipping from rest down an inclined plane. What shape should a body have to reach the bottom of the incline last? Does it depend on the mass of the object? Does it depend on the shape (radius) of the object?

Or both?

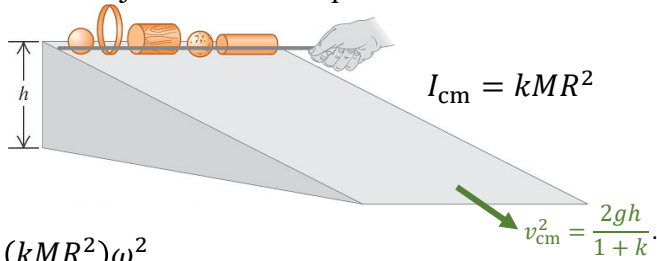
Answer:

By conservation of total mechanical energy,

$$Mgh = \frac{1}{2} Mv_{\text{cm}}^2 + \frac{1}{2} (kMR^2) \omega^2$$

Since the bodies **roll without slipping**, $v_{\text{cm}} = R\omega$.

We can prove that $v_{\text{cm}}^2 = \frac{2gh}{1+k}$.



The ring will be the last one to reach the bottom regardless of its mass and radius!!

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Question. Prove that for any two solid spheres of different radius and mass rolling from rest from an inclined plane, their time to travel from the top of the plane to the bottom will be the same. **Compute the time if the initial height is h and the angle of inclined plane is θ .** Assume the motion to be pure rolling.

Answer:

The total kinetic energy of the solid sphere on the bottom of slope is

$$\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2.$$

$$v = r\omega \text{ (for pure rolling)} \rightarrow \omega = \frac{v}{r}.$$

As the total energy is conserved, we

$$\text{have } \frac{1}{2}mv^2 + \frac{1}{2}I\left(\frac{v}{r}\right)^2 = mgh.$$

$$\frac{1}{2}\left(m + \frac{I}{r^2}\right)v^2 = mgh$$

$$\rightarrow v^2 = \frac{2gh}{1 + \frac{I}{mr^2}}$$

$$I_{\text{solid sphere}} = \frac{2}{5}mr^2$$

$$\rightarrow v^2 = \frac{2gh}{1 + \frac{2mr^2}{5mr^2}} = \frac{2gh}{7/5} = \frac{10gh}{7}$$

$$\rightarrow v = \sqrt{\frac{10gh}{7}}$$

$$\text{From } \Delta s = \left(\frac{v_i + v_f}{2}\right)t$$

$$\rightarrow \frac{h}{\sin \theta} = \left(\frac{0 + \sqrt{\frac{10gh}{7}}}{2}\right)t$$

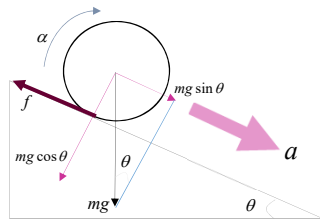
$$\rightarrow t = \frac{h}{\sin \theta} \left(\frac{\sqrt{28}}{\sqrt{10gh}}\right) = \sqrt{\frac{28h}{10g\sin^2 \theta}} = \sqrt{\frac{14h}{5g\sin^2 \theta}}$$

As the arrival time at the bottom is only dependent on the initial height (the gravitational acceleration and the inclined angle are constant), both solid spheres will reach the bottom at the same time **regardless of their radius and mass.**

Analysis by Energy

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Alternatively, you can compute the
acceleration from the elementary
model



$$mg \sin \theta - f = ma$$

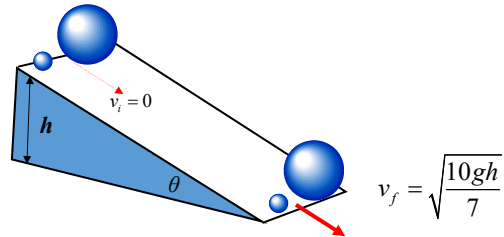
$$\rightarrow f = mg \sin \theta - ma$$

$$f \times r = I\alpha \quad a = r\alpha \text{ (for pure rolling)} \rightarrow \alpha = \frac{a}{r}$$

$$\rightarrow (mg \sin \theta - ma) \times r = \left(\frac{2}{5}mr^2\right)\left(\frac{a}{r}\right)$$

$$g \sin \theta - a = \frac{2a}{5}$$

$$a = \frac{5g \sin \theta}{7} < g \sin \theta$$



Analysis by Newton's 2nd Law

$$\text{From } v_f = v_i + at$$

$$t = \frac{v_f}{a} = \frac{\sqrt{\frac{10gh}{7}}}{\frac{5g \sin \theta}{7}} = \left(\sqrt{\frac{10gh}{7}}\right)\left(\frac{7}{5g \sin \theta}\right)$$

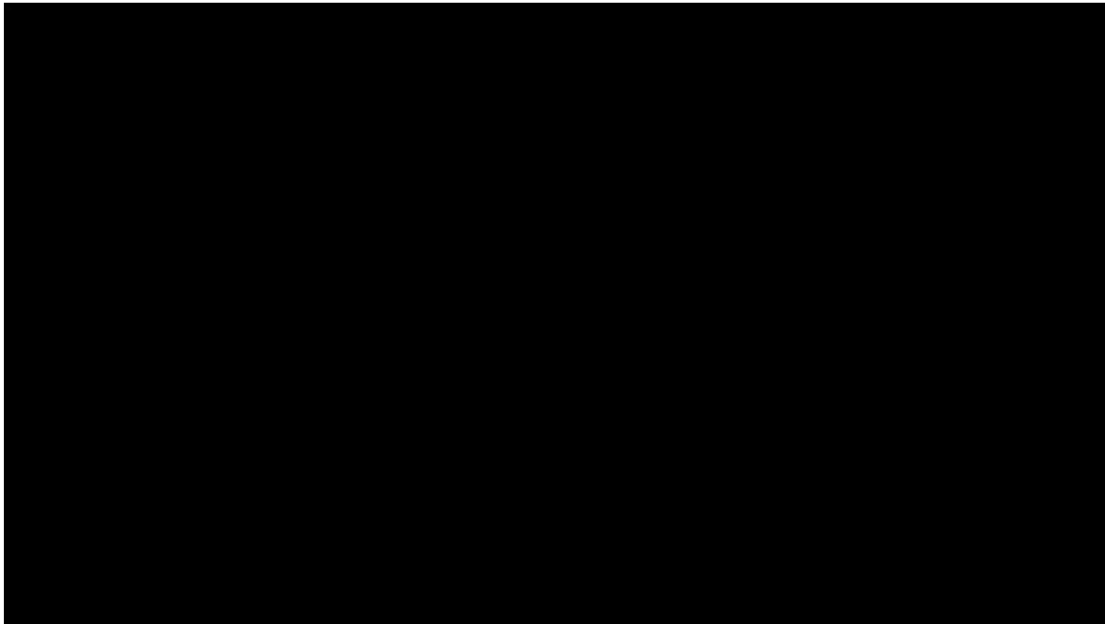
$$= \sqrt{\frac{2 \cdot 10gh \times 7 \times 7}{7 \times 5 \times 5 \times g \times \sin^2 \theta}}$$

$$t = \sqrt{\frac{14h}{5g \sin^2 \theta}}$$

Same answer.

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Effect of Moment of Inertia to Linear Acceleration

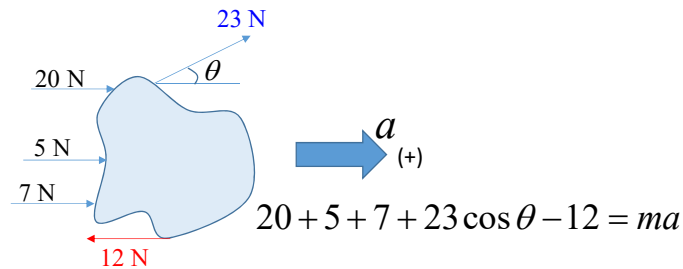


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Newton's Second Law

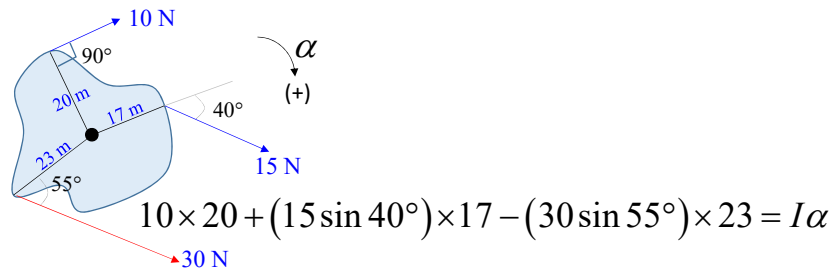
Translation

$$\sum F_i = ma$$



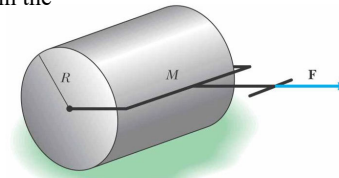
Rotation

$$\sum \tau_i = I\alpha$$



Problem. A constant horizontal force $\mathbf{F}$ is applied to a lawn roller in the form of a uniform solid cylinder of radius R and mass M . If the roller rolls without slipping on the horizontal surface, find

- the acceleration of the center of mass and
- the minimum coefficient of static friction necessary to prevent slipping.



Answer:

(i) Linear motion: $F - f = Ma \dots (1)$

Rotational motion: $f \times R = I\alpha \dots (2)$

Pure rolling: $a = R\alpha \dots (3)$

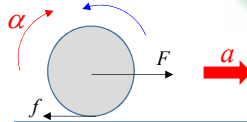
Moment of inertia: $I = \frac{MR^2}{2} \dots (4)$

The rest is algebra!!

$$a = \frac{F - f}{M} = \frac{F - \frac{I\alpha}{R}}{M} = \frac{F - \frac{I}{R} \left(\frac{a}{R} \right)}{M}$$

$$\rightarrow aM = F - \frac{MR^2}{2} \left(\frac{a}{R} \right) \rightarrow aM = F - \frac{Ma}{2}$$

$$\rightarrow \frac{3Ma}{2} = F \rightarrow a = \frac{2F}{3M}$$



- (ii) When there is no slipping,

$$f_s = \mu_s Mg \quad (\text{it does not rub!!})$$

$$\text{Torque} = (\mu_s Mg) \times R = I\alpha$$

$$\rightarrow (\mu_s Mg) \times R = \left(\frac{MR^2}{2} \right) \left(\frac{a}{R} \right)$$

$$\rightarrow (\mu_s Mg) \times R = \frac{1}{2} MRa = \frac{1}{2} MR \left(\frac{2F}{3M} \right)$$

$$\rightarrow \mu_{\text{minimum}} = \frac{F}{3Mg}$$